



kingdom of Saudi Arabia
Ministry of Education
Hafr Al Baten Directorate for boys
Al –Ajial International Schools



Student's Name		Grade	10
Subject	Computer	worksheet	

Q1: choose the correct answer:

1-What is the primary goal of supervised learning?

- a) To discover hidden patterns in unlabeled data.
- b) To predict an output variable based on input features using labeled data. ☒
- c) To reduce the dimensionality of data.
- d) To optimize a reward function through trial and error.

2-Which of the following is an example of a classification algorithm?

- a) Linear Regression
- b) K-Means Clustering
- c) Decision Trees ☒
- d) Principal Component Analysis (PCA)

3-What does the term "overfitting" mean in ML?

- a) The model performs well on training data but poorly on unseen data. ☒
- b) The model is too simple to capture patterns in the data.
- c) The model has high bias.
- d) The model ignores noise in the data.

4-Which evaluation metric is most suitable for imbalanced classification problems?

- a) Accuracy
- b) Mean Squared Error (MSE)
- c) F1-Score ☒
- d) R-Squared



Student's Name		Grade	10
Subject	Computer	worksheet	

Q1: Compare and contrast decision trees and logistic regression:

- **Decision Trees:**

- Non-linear decision boundaries
- Can handle mixed feature types
- Interpretable as rules
- Prone to overfitting
- No need for feature scaling

- **Logistic Regression:**

- Linear decision boundary (without transformations)
- Requires numerical features
- Probabilistic interpretation
- Less prone to overfitting with regularization
- Requires feature scaling for best performance

Short Answer Questions:

*Why do we split data into training and test sets?

Answer: To evaluate how well the model generalizes to unseen data and detect overfitting.



kingdom of Saudi Arabia
Ministry of Education
Hafr Al Baten Directorate for boys
Al –Ajial International Schools



Student's Name		Grade	10
Subject	Computer	worksheet	

Q1: choose the correct answer:

1- What is the first step in character modeling?

Rigging	Sketching	Texturing	Animation
---------	------------------	-----------	-----------

2- Which of the following is NOT a 3D modeling technique?

Box Modeling	Sculpting	Keyframing	Edge Looping
--------------	-----------	-------------------	--------------

3- Which software is primarily used for character animation?

Adobe Photoshop	Microsoft Word	Microsoft Excel	Autodesk Maya
-----------------	----------------	-----------------	----------------------

4- A _____ is a sequence of keyframes that create movement.

Animation	Unity	ARKit	Unreal Engine
------------------	-------	-------	---------------

Q2: Read the following and put (✓) for the correct answer and (x) for the wrong answer

1) Subdivision surfaces are used to reduce polygon count in a model.	(x)
2) Inverse Kinematics (IK) helps in animating limbs more naturally.	(✓)
3) UV mapping is the process of animating a character.	(x)
4) Keyframes define the start and end points of an animation.	(✓)



Student's Name		Grade	10
----------------	-------	-------	----

Q1 : Read and put True(✓) or false (✗) :

- 1- **Supervised learning** requires labeled data for training. **(True)**
- 2- **Unsupervised learning** deals with unlabeled data. **(True)**
- 3- **Overfitting** occurs when a model performs well on training data but poorly on unseen data. **(True)**
- 4- **Classification** is a type of regression problem. **(False)**
- 5- **Linear regression** is used for predicting continuous values. **(True)**
- 6- **Decision trees** can only be used for classification tasks. **(False)**
- 7- **K-means clustering** is a supervised learning algorithm. **(False)**
- 8- **Feature scaling** is essential for algorithms like SVM and KNN. **(True)**
- 9- **Deep learning** is a subset of reinforcement learning. **(False)**
- 10- **Bias** in a model refers to the error introduced by approximating real-world problems. **(True)**
- 11- **Cross-validation** helps in reducing overfitting. **(True)**
- 12- **PCA (Principal Component Analysis)** is used for feature extraction. **(True)**
- 13- **Reinforcement learning** relies solely on historical data without any interaction. **(False)**
- 14- **A confusion matrix** is used to evaluate classification models. **(True)**
- 15- **Gradient descent** is used to minimize the loss function. **(True)**



Student's Name

.....

Grade

10

Q3(A) Match the word in column (A) with the correct option
in column (B):

Attribute	#	value
1- Mesh	2	The process of adding bones to a model
2- Rigging	1	A 3D object made of polygons
3- Keyframe	3	A pose saved at a specific time
4- Morph Target	5	Simulates cloth movement
5- Cloth Simulation	4	Shape key for facial expressions

(B) Re-arrange the next stages to Character Modeling Process:

- (3) UV Unwrapping
(2) Sculpting/Modeling
(5) Rigging
(4) Texturing
(1) Concept Art



kingdom of Saudi Arabia
Ministry of Education
Hafr Al Baten Directorate for boys
Al –Ajial International Schools



Student's Name		Grade	10
----------------	-------	-------	----

Q4: complete the following from between brackets:

(**Vertex Animation** - **Texturing** - **Aim** - **Extrusion or Modeling**)

- 1- The input variables in a dataset are called Extrusion or Modeling.
- 2- The Aim constraint in rigging forces a bone to always point at a target.
- 3- Adding colors and patterns to a 3D model is called Texturing.
- 4- The Vertex Animation animation technique involves moving vertices manually.